

Week 12 Day 2: Dataset Sources and Data Dictionaries

Today we connect the final project to real public datasets and learn how to document what a dataset means.

Today

- find a dataset
- read its structure
- build a mini data dictionary

Why

data analysis starts with understanding columns, rows, and meanings

Output

- a dataset choice
- a data dictionary
- a final project connection

Today continues data analytics for security

Last class

- security metrics
- simple scoring
- final project paths
- coding or MLA essay

Today

- finding a dataset
- reading rows and columns
- building a data dictionary
- documenting concerns

Several public dataset sources can work for this lesson

Civic data

NYC Open Data
Data.gov
complaints, inspections,
infrastructure

Security data

CISA vulnerability catalog
NIST vulnerability database
vendors, products, software
flaws

Privacy data

HHS health breach portal
Kaggle optional
breaches, sensitive data,
public datasets

Good dataset searches start with course themes

security: alerts, systems, infrastructure, inspections

privacy: location, time, identity, sensitive categories

ethics: fairness, missing data, public impact

cloud and databases: records, fields, access, retention

analytics: counts, categories, trends, comparisons

A dataset is usually a table of rows, columns, and values

Row

one record
one event
one system
one complaint
one inspection

Column

one field
one attribute
same kind of information
for every row

Value

one cell entry
text, number, date,
category, location

A data dictionary explains what the columns mean

Without one

column names can be abbreviated
codes may be unclear
missing values may be invisible
students may guess wrong

With one

columns have plain English meanings
data types are clearer
example values make sense
concerns become visible

A useful mini data dictionary has six parts

Column name: copy the exact field name.

Plain English meaning: explain the field normally.

Data type: text, number, date, category, location, yes/no.

Example value: copy one realistic value.

Missing or unclear: note blanks, codes, or confusion.

Concern: name a security, privacy, or ethics issue.

Example

The same column can be useful and risky

Useful for analysis

- `incident\_date` can show patterns over time.
- `borough` can support geographic comparison.
- `status` can show unresolved records.
- `category` can group similar records.

Possible concerns

- time plus location can reveal patterns
- categories may be biased or unclear
- missing values can distort conclusions
- neighborhood comparisons can be misused

Security questions focus on systems, access, and misuse

Ask

Could this reveal system weakness?

Could it expose infrastructure?

Could it guide misuse?

Look for

system names

locations

status fields

incident details

Document

what is useful

what is sensitive

what should be handled carefully

Privacy questions focus on people and linkability

Direct

names
addresses
IDs
contact fields

Indirect

time
location
small groups
rare categories

Linked

combined columns
merged datasets
patterns across records

Ethics questions focus on interpretation and harm

Common risks

blaming neighborhoods without context
treating missing data as neutral
confusing reports with confirmed facts
ignoring who benefits from the analysis

Responsible habits

state limitations
avoid overclaiming
explain uncertainty
connect claims to evidence

Today's in-class lab has six parts

1. Choose one dataset source and one dataset.
2. Take a first look at rows, columns, dates, and locations.
3. Build a mini data dictionary for 6-8 columns.
4. Document security, privacy, and ethics concerns.
5. Connect the dataset to your final project path.
6. Write a short reflection before submitting.

How a dataset can support the coding project

Simple code actions

filter rows
count values
group by category
sort by date or count
make a summary table

Analyst-style output

what changed?
what stands out?
what should be reviewed?
what concern or limitation matters?

How a dataset can support the research essay

Dataset as evidence

- shows a real-world pattern
- illustrates a public issue
- raises a privacy question
- shows limits of categories

Other sources

- course reading
- official documentation
- credible reporting
- government or standards source

In-Class Finish

Use the last block to complete and submit the dataset lab

Finish

dataset choice
6-8 column data dictionary
security/privacy/ethics
concerns

Write

120-180 word reflection
what you understand better
one concern you noticed

Submit

one document
before leaving class
no homework today

End of Day 2

A dataset is not just something you download. It is something you have to understand before you use.

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